# Population -> sample -> statistic -> sampling distribution



## 1 RECOGNISE IT

- Repeated random samples of the same size are imagined.
- The random object is a statistic such as $\bar{X}$ or p.
- Asked about how a statistic varies across samples.

## 2 DECISION / METHOD

Population (μ,σ,π) -> draw sample of size n -> compute statistic ($\bar{X}$,p) -> repeat conceptually -> distribution of statistic.

## 3 FORMULA

```
population variable: X
sample statistic: X̄ or p
sampling distribution: all possible
statistic values for fixed n
```

## 4 FORMULA SHEET VS WHAT I KNOW

Gives: parameter/statistic pairings and their standard errors under conditions.
Still know: the four-stage chain and which random object the probability concerns.

## 5 EXCEL ROUTE

Use Excel only after naming the object:
- raw sample mean: `=AVERAGE(range)`
- sample size: `=COUNT(range)`
- probability model comes from the sampling distribution.

## 6 TINY WORKED EXAMPLE

Population values generate many samples. If each sample has n=25, every sample gives a different $\bar{X}$; those $\bar{X}$ values form the sampling distribution.

## 7 TOP EXAM TRAPS

- Calling the sample histogram the sampling distribution.
- Using raw-X SD when the event is about $\bar{X}$.
- Confusing parameter μ with statistic $\bar{X}$.

## 8 SAY ALOUD

A sampling distribution is the distribution of a statistic over all possible samples of one fixed size.

## 9 EXIT CHECK

In $P(\bar{X}>10)$, what is random: one observation or the sample mean?

**ONE-GLANCE VISUAL ROUTE**

